

Problem-Solution Fit

HDI Prediction System — Project Design Phase

Problem Statement

Human Development Index (HDI) is a composite measure used globally to compare the well-being and development level of countries. Governments, researchers, and policy analysts need a fast, data-driven way to estimate or validate HDI values using core socio-economic indicators, without waiting for the full annual UNDP report cycle or performing manual index computation.

Problems Identified

- HDI computation is not readily accessible as an interactive, on-demand tool for students, researchers, or planners.
- Existing HDI data is published periodically (annually), making it hard to explore what-if scenarios for individual countries or regions.
- Manually computing HDI from raw indicators (life expectancy, schooling years, GNI per capita) requires statistical knowledge most users don't have.
- There is no lightweight, deployable web tool that predicts HDI directly from these four core indicators using a trained machine learning model.

Target Users

- Students and researchers studying development economics or public policy.
- Data science learners exploring regression modelling on real-world socio-economic datasets.
- Policy analysts who want a quick, indicative HDI estimate for a given set of indicators.

Solution Fit

A supervised machine learning model — trained on 2021 global HDI data covering 195 countries — can learn the relationship between the four core indicators (Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita) and the published HDI value. Wrapping this model in a simple, web-based interface allows any user to enter indicator values and instantly receive a predicted HDI score, closing the gap between raw statistics and usable insight.

- Linear Regression was selected for its interpretability and strong fit on this dataset (R^2 approx. 0.98).
- A Flask-based web application exposes the model through a simple form-based UI.
- The solution is deployed on Render.com, making it publicly accessible without local setup.

Why This Fit Works

The strong correlation between the four chosen indicators and HDI (validated by the high R^2 score during training) confirms that a lightweight regression-based approach is sufficient — avoiding the complexity of deep learning while still delivering accurate, real-time predictions suited to the target users' needs.